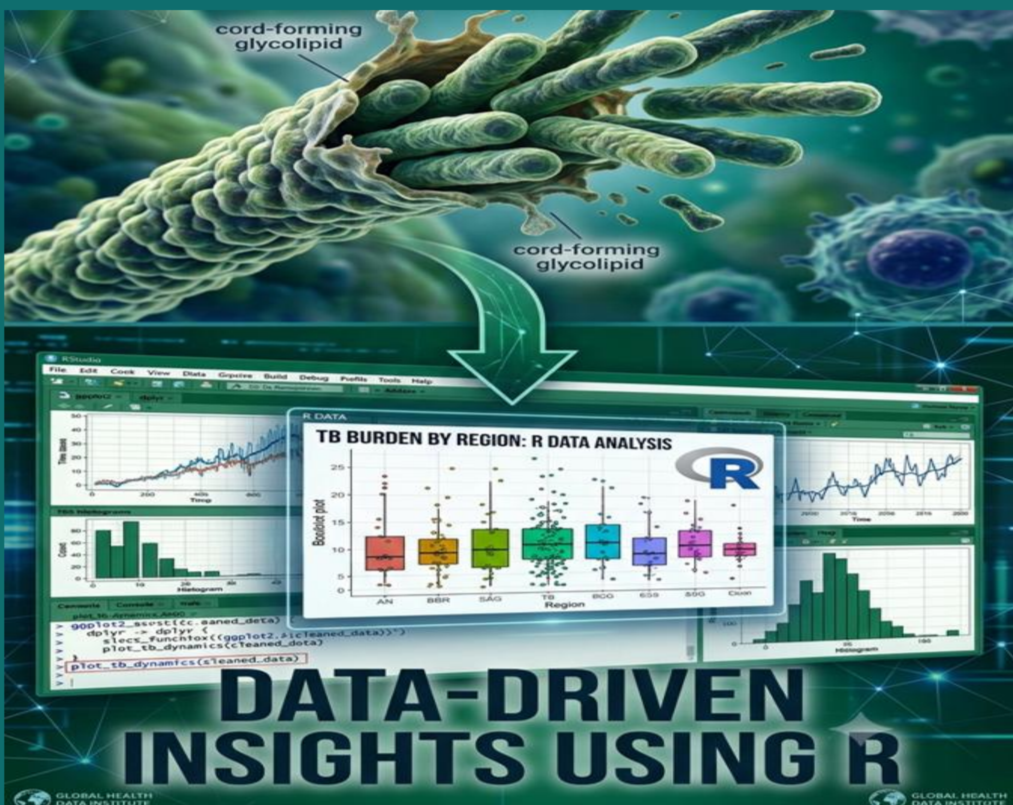
# CAPSTONE

# PROJECT

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An assignment of R classes by Dr. Aswath Karunakaran

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## Background

In this project, we examine clinical data from TB patients to test whether medical officers appropriately assess risk based on common danger signs and symptoms, or whether an algorithmic (rule-based) approach to risk stratification would perform better — particularly for identifying seriously ill patients who need to be admitted or referred urgently. The analysis below is organized question by question, following the study brief.

## Question 1: Bring the Date of Assessment Column First and Rename it to "start"

**Step 1 — Rename and relocate the column.**

```
data <- data %>%
  rename(start = date_of_assessment) %>%
  relocate(start, .before = end)

head(data)
```

```
# A tibble: 6 × 57
  start               end                 district episode_id patient_name
age
  <dttm>              <dttm>              <chr>    <lgl>      <lgl>        <d
bl>
1 2022-10-20 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
33
2 2022-10-17 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
60
3 2022-10-19 00:00:00 2022-10-30 00:00:00 Baraban… NA         NA
30
4 2022-10-15 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
20
5 2022-10-18 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
35
6 2022-10-17 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
45
# ℹ 51 more variables: gender <chr>,
#   does_the_patient_reside_in_the_same_district_as_the_current_implementing_
facility_s_district <chr>,
#   name_of_other_district <chr>, current_symptoms <chr>,
```

```
#   current_symptoms_none <dbl>, current_symptoms_cough <dbl>,
#   current_symptoms_persistent_fever <dbl>,
#   current_symptoms_evening_rise_of_temperature <dbl>,
#   current_symptoms_breathlessness <dbl>, blood_in_sputum <dbl>, …
```

**Result:** The `date_of_assessment` column is renamed to `start` and moved to the first position in the dataset, ahead of the end column.

## Question 2: Create a Unique ID from the Row ID

**Step 1 — Generate the ID column.**

```
data <- rownames_to_column(data, var = "id")
head(data$id)

[1] "1" "2" "3" "4" "5" "6"
```

**Result:** Each patient record now has a unique `id` derived from its row number.

## Question 3: Mean Delay in Assessment and Mean Age

**Step 1 — Calculate assessment delay.**

```
data <- data %>%
  mutate(assessment_delay = as.numeric(difftime(end, start, units = "days")))
```

**Step 2 — Calculate the means.**

```
mean_delay <- mean(data$assessment_delay, na.rm = TRUE)
mean_age   <- mean(data$age, na.rm = TRUE)

mean_delay

[1] 8.26281

mean_age

[1] 32.27333
```

**Step 3 — Check for missing values.**

```
sum(is.na(data$assessment_delay))

[1] 0

sum(is.na(data$age))
```

```
[1] 5
```

**Step 4 — Replace missing age values with the mean.**

```
data <- data %>%
  mutate(age = ifelse(is.na(age), mean_age, age))
```

**Result:** The mean delay in assessment was 8.26 days, and the mean age of patients was 32.27 years. Missing age values were replaced with the mean age.

## Question 4: Percentage of Patients with Cough as a Symptom

**Step 1 — Calculate the percentage.**

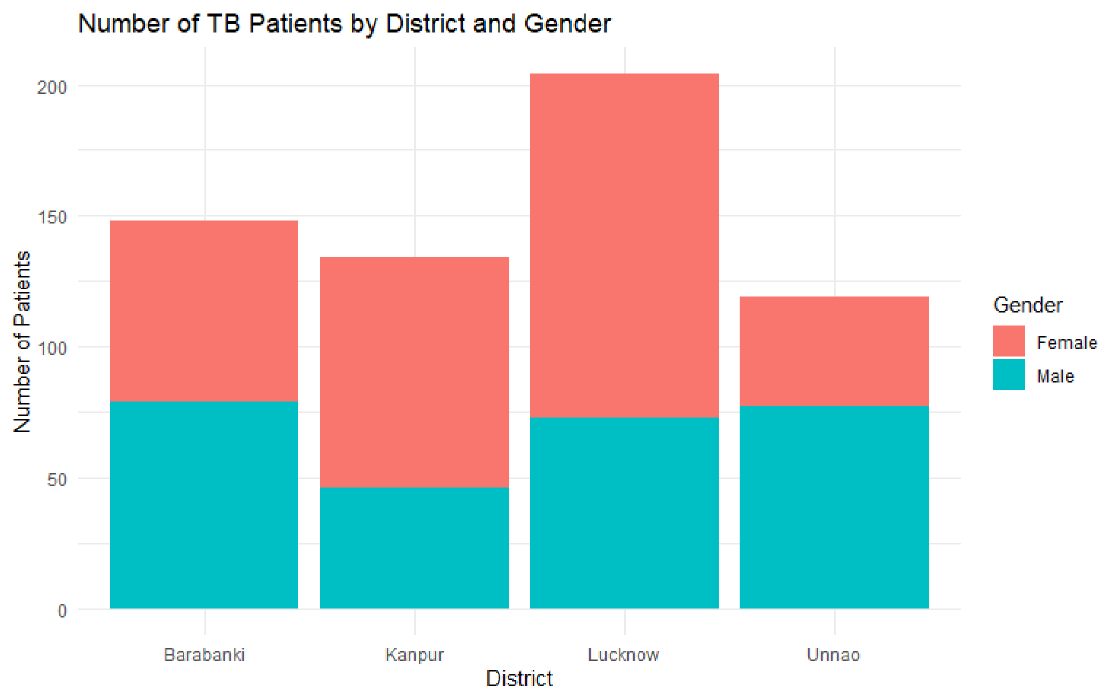
```
cough_percentage <- mean(data$current_symptoms_cough) * 100
round(cough_percentage, 2)

[1] 58.02
```

**Result:** 58.02% of patients reported cough as a symptom.

## Question 5: Stacked Column Chart of Patients by District and Gender

**Step 1 — Build the chart.**

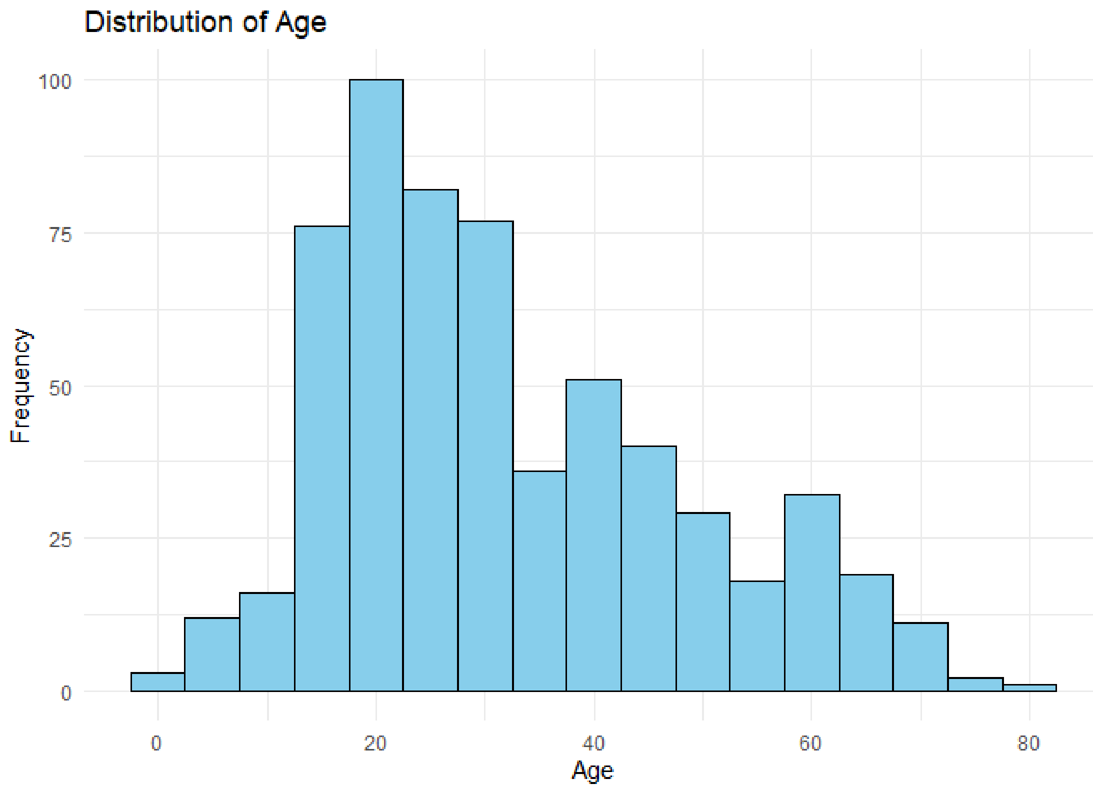
```
ggplot(data, aes(x = district, fill = gender)) +
  geom_bar() +
  labs(
    title = "Number of TB Patients by District and Gender",
    x = "District",
    y = "Number of Patients",
    fill = "Gender"
  ) +
  theme_minimal()
```




## Question 6: Does Age Follow a Normal Distribution?

**Step 1 — Visualize the distribution.**

```
ggplot(data, aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Age",
    x = "Age",
    y = "Frequency"
  ) +
  theme_minimal()
```




**Step 2 — Confirm statistically with the Shapiro-Wilk test.**

```
shapiro.test(data$age)


    Shapiro-Wilk normality test

data:  data$age
W = 0.94581, p-value = 4.495e-14
```

**Interpretation:**

- The age distribution is not symmetric.
- Most patients fall between 15 and 35 years.
- There is a long right tail extending to older ages (50–80 years), indicating a **positively (right) skewed** distribution rather than a bell-shaped one.
- The Shapiro-Wilk test confirms this statistically ($W = 0.946$, $p < 0.001$).

**Conclusion:** Age does **not** follow a normal distribution.

## Question 7: Stratify Patients by Number of Abnormal Parameters

Patients are stratified into risk categories based on the number of abnormal clinical parameters:

- **1 abnormal parameter** → Low Risk
- **2–3 abnormal parameters** → Moderate Risk
- **≥4 abnormal parameters** → High Risk

**Step 1 — Create the risk category variable.**

```
data <- data %>%
  mutate(
    risk_category = case_when(
      no_of_abnormal_parameters == 1 ~ "Low Risk",
      no_of_abnormal_parameters >= 2 & no_of_abnormal_parameters <= 3 ~ "Mode
rate Risk",
      no_of_abnormal_parameters >= 4 ~ "High Risk",
      TRUE ~ NA_character_
    )
  )

table(data$risk_category)


    High Risk      Low Risk Moderate Risk
          226            91           258
```

## Question 8: Cross-Tabulate Medical Officer Risk vs. Algorithmic Risk

This is the central question of the project: how often does a medical officer under-classify a patient as low risk when the objective parameter count places them at high risk?

**Step 1 — Clean up the medical officer's risk labels.**

```
data$mo_risk <- recode(
  data$final_risk_stratification_by_the_medical_officer,
  "Low risk- Marked for provisional follow-up later." = "Low Risk",
  "Moderate Risk- Marked for mandatory follow-up later at CHC/PHC/HWC" = "Mod
erate Risk",
  "High risk- Referring to District Hospital/Sub-district hospital or tertiar
y care facility with availability of intensive or emergency or inpatient care
```

```
." = "High Risk"
)
```

**Step 2 — Cross-tabulate.**

```
cross_tab <- table(data$mo_risk, data$risk_category)
cross_tab


                High Risk Low Risk Moderate Risk
  High Risk             9        0             3
  Low Risk            185       87           230
  Moderate Risk        30        2            18
```

**Interpretation:** The cross-tabulation above shows the number of patients whom the medical officer classified as low risk, cross-referenced against their algorithmic (parameter-count-based) risk category. Any patients falling in the "Low Risk" row but "High Risk" column represent cases where the medical officer under-assessed a seriously ill patient — the key safety concern this project set out to investigate.

## Question 9: Descending Order Chart of Abnormal Parameters

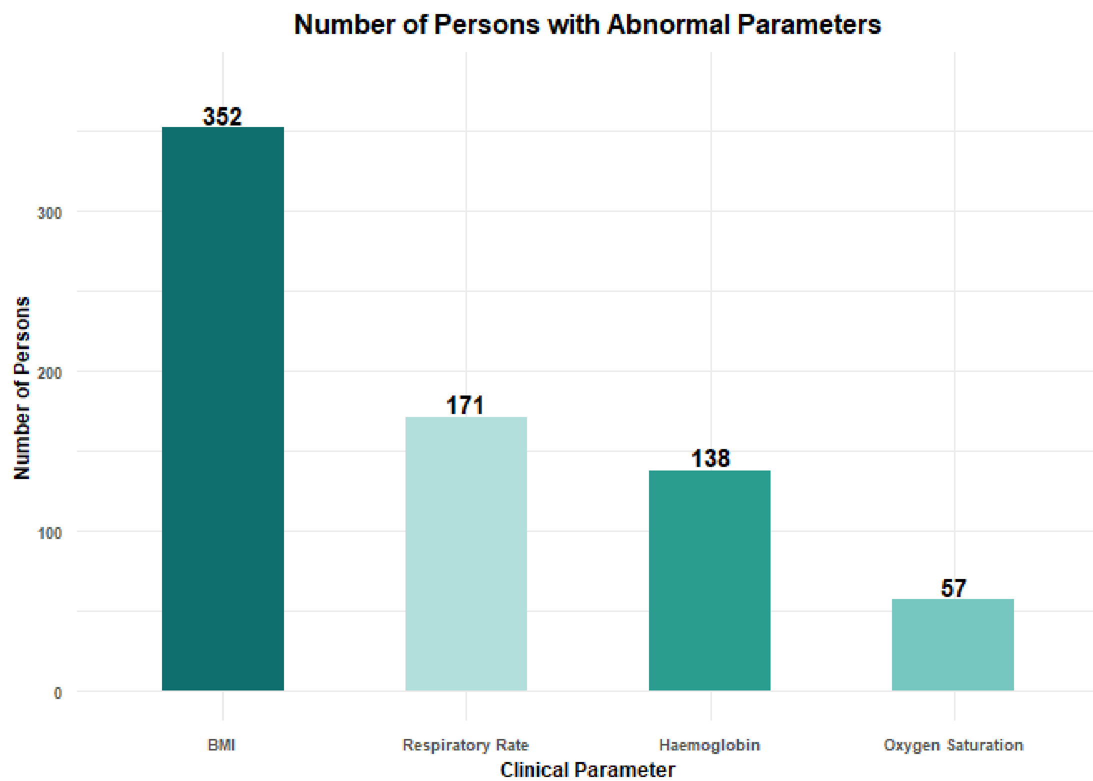
**Step 1 — Count patients with each abnormal parameter.**

```
# Each of these columns is a 0/1 flag per the data dictionary (1 = abnormal),
# so column sums give the number of patients with that parameter abnormal.
abnormal_counts <- data.frame(
  Parameter = c("Respiratory Rate", "BMI", "Haemoglobin", "Oxygen Saturation"
),
  Count = c(
    sum(data$respiratory_rate, na.rm = TRUE),
    sum(data$bmi, na.rm = TRUE),
    sum(data$haemoglobin, na.rm = TRUE),
    sum(data$oxygen_saturation, na.rm = TRUE)
  )
) %>%
  arrange(desc(Count))

abnormal_counts

          Parameter Count
1               BMI   352
2  Respiratory Rate   171
```

```
3       Haemoglobin   138
4 Oxygen Saturation    57
```

**Step 2 — Plot in descending order.**

```
ggplot(abnormal_counts,
       aes(x = reorder(Parameter, -Count), y = Count, fill = Parameter)) +
  geom_col(width = 0.5) +
  geom_text(aes(label = Count), vjust = -0.2, size = 4, fontface = "bold") +
  scale_fill_manual(values = c("#0F6E6E", "#2A9D8F", "#76C7C0", "#B2DFDB")) +
  labs(
    title = "Number of Persons with Abnormal Parameters",
    x = "Clinical Parameter",
    y = "Number of Persons"
  ) +
  expand_limits(y = 380) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 9),
    axis.text = element_text(face = "bold", size = 7),
    legend.position = "none"
  )
```




**Result:** BMI was the most frequently abnormal parameter (352 patients), followed by respiratory rate (171), haemoglobin (138), and oxygen saturation (57).

## Question 10: Subset of Male Patients Without Blood in Sputum

**Step 1 — Create the subset.**

```
male_no_blood <- subset(data, gender == "Male" & blood_in_sputum == 0)
nrow(male_no_blood)
```

```
[1] 269
```

```
head(male_no_blood)
```

```
# A tibble: 6 × 61
  id    start               end                 district episode_id patient_n
ame
  <chr> <dttm>              <dttm>              <chr>    <lgl>      <lgl>
1 4     2022-10-15 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
2 6     2022-10-17 00:00:00 2022-10-29 00:00:00 Baraban… NA         NA
3 9     2022-10-22 00:00:00 2022-10-30 00:00:00 Baraban… NA         NA
4 12    2022-10-15 00:00:00 2022-10-30 00:00:00 Kanpur   NA         NA
```

```
5 19    2022-10-20 00:00:00 2022-10-30 00:00:00 Kanpur   NA         NA
6 23    2022-10-28 00:00:00 2022-10-31 00:00:00 Kanpur   NA         NA
# ℹ 55 more variables: age <dbl>, gender <chr>,
#   does_the_patient_reside_in_the_same_district_as_the_current_implementing_
facility_s_district <chr>,
#   name_of_other_district <chr>, current_symptoms <chr>,
#   current_symptoms_none <dbl>, current_symptoms_cough <dbl>,
#   current_symptoms_persistent_fever <dbl>,
#   current_symptoms_evening_rise_of_temperature <dbl>,
#   current_symptoms_breathlessness <dbl>, blood_in_sputum <dbl>, …
```

## Question 11: Additional Analysis (Optional)

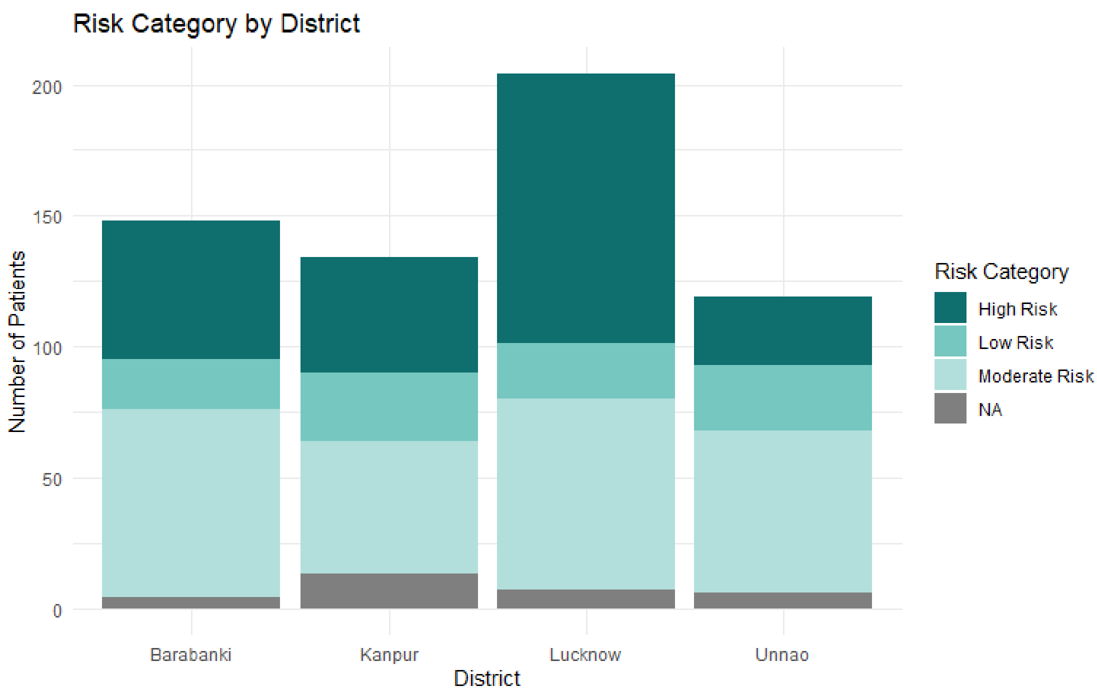
*Exploratory analysis beyond the core questions.*

### Step 1: Risk Category by District

```r
table(data$district, data$risk_category)
```

```
           High Risk Low Risk Moderate Risk
  Barabanki        53       19            72
  Kanpur           44       26            51
  Lucknow         103       21            73
  Unnao            26       25            62
```

```r
ggplot(data, aes(x = district, fill = risk_category)) +
  geom_bar() +
  scale_fill_manual(values = c("#0F6E6E", "#76C7C0", "#B2DFDB")) +
  labs(
    title = "Risk Category by District",
    x = "District",
    y = "Number of Patients",
    fill = "Risk Category"
  ) +
  theme_minimal()
```



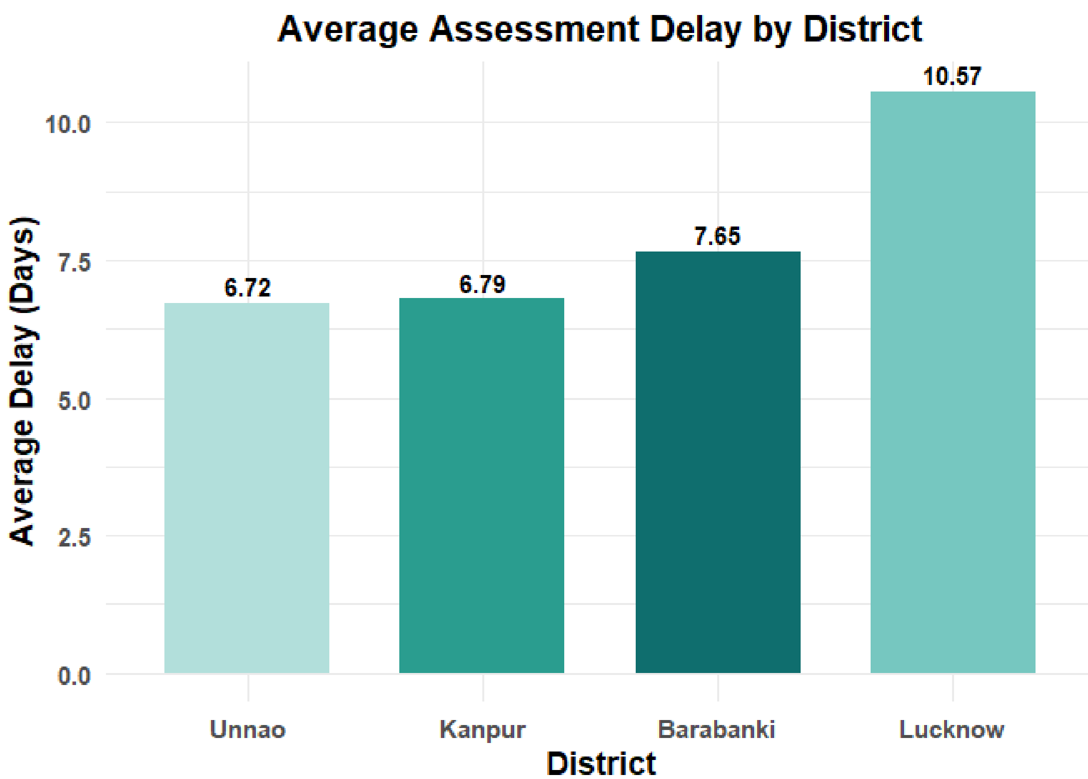

## Step 2: Assessment Delay by District

```
delay_by_district <- aggregate(
  assessment_delay ~ district,
  data = data,
  mean,
  na.rm = TRUE
)
delay_by_district$assessment_delay <- round(delay_by_district$assessment_dela
y, 2)
delay_by_district
```

```
   district assessment_delay
1 Barabanki             7.65
2    Kanpur             6.79
3   Lucknow            10.57
4     Unnao             6.72
```

```
ggplot(delay_by_district,
       aes(x = reorder(district, assessment_delay), y = assessment_delay, fil
l = district)) +
  geom_col(width = 0.7) +
  scale_fill_manual(values = c("#0F6E6E", "#2A9D8F", "#76C7C0", "#B2DFDB")) +
  geom_text(aes(label = assessment_delay), vjust = -0.4, fontface = "bold", s
ize = 4) +
```

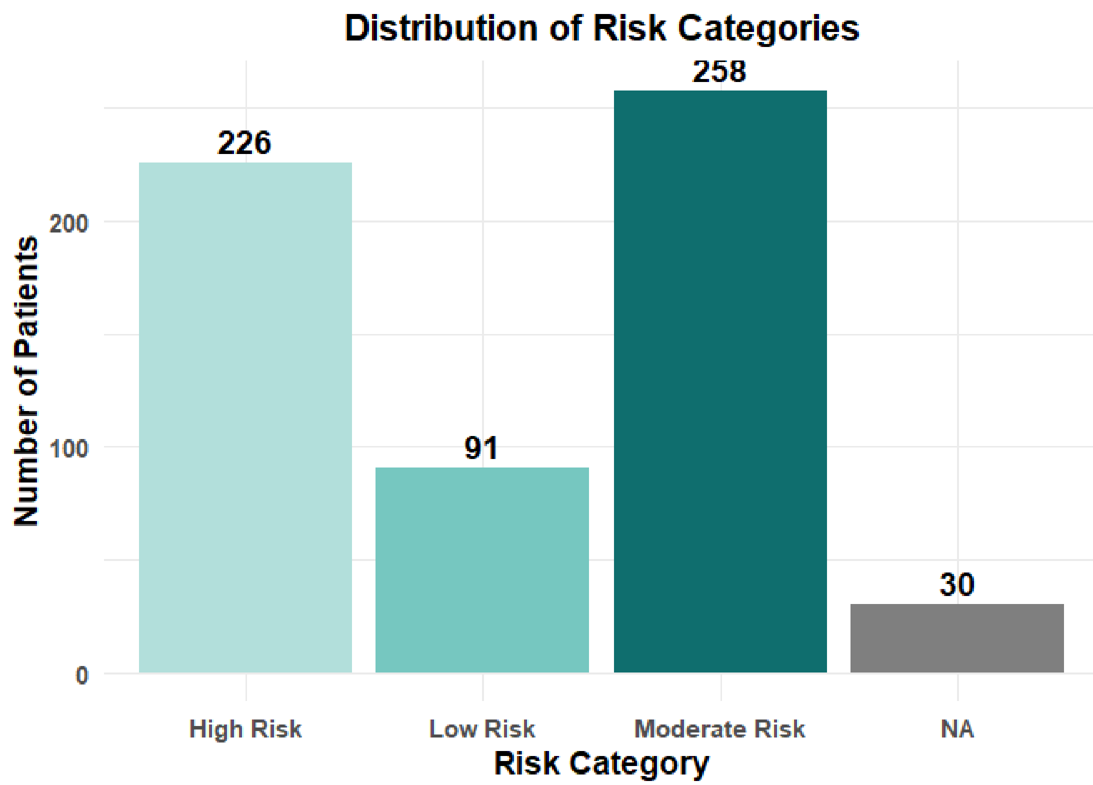
```
  labs(
    title = "Average Assessment Delay by District",
    x = "District",
    y = "Average Delay (Days)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(face = "bold")
  )
```




## Step 3: Distribution of Risk Categories

```
ggplot(data, aes(risk_category, fill = risk_category)) +
  geom_bar() +
  scale_fill_manual(values = c("#B2DFDB", "#76C7C0", "#0F6E6E")) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.4, fon
tface = "bold") +
  labs(
    title = "Distribution of Risk Categories",
```

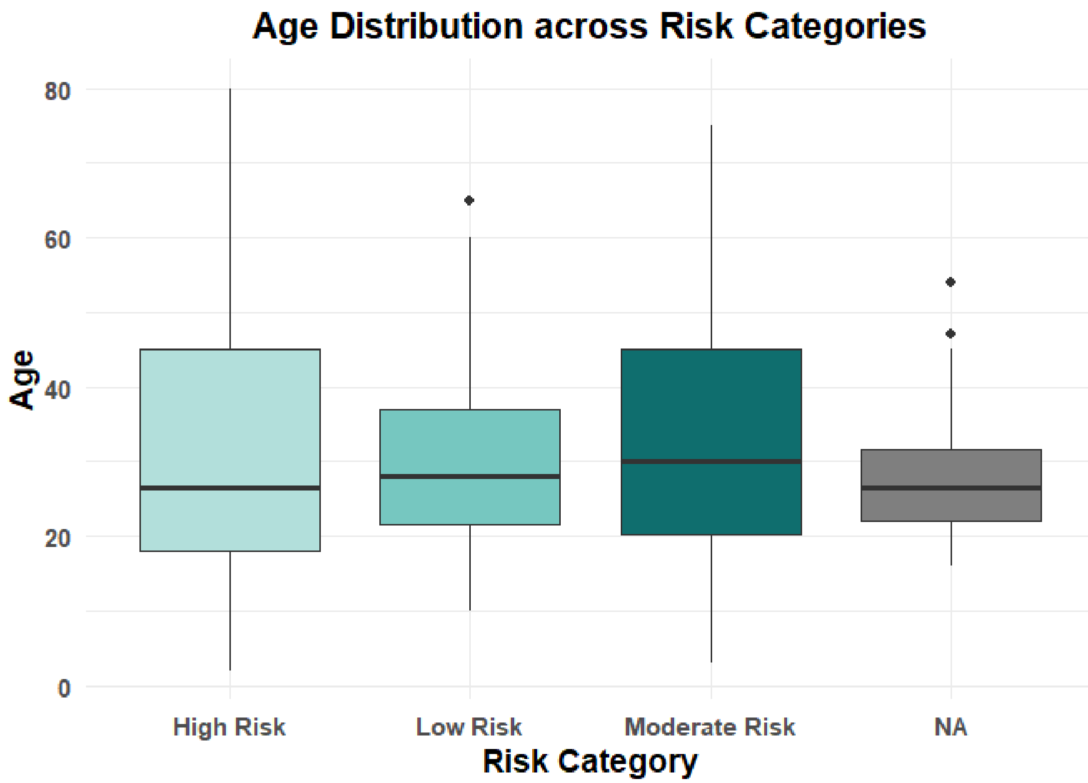
```r
    x = "Risk Category",
    y = "Number of Patients"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(face = "bold")
  )
```




### Step 4: Age Distribution by Risk Category

```r
ggplot(data, aes(risk_category, age, fill = risk_category)) +
  geom_boxplot() +
  scale_fill_manual(values = c("#B2DFDB", "#76C7C0", "#0F6E6E")) +
  labs(
    title = "Age Distribution across Risk Categories",
    x = "Risk Category",
    y = "Age"
  ) +
  theme_minimal(base_size = 14) +
```

```r
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(face = "bold")
  )
```

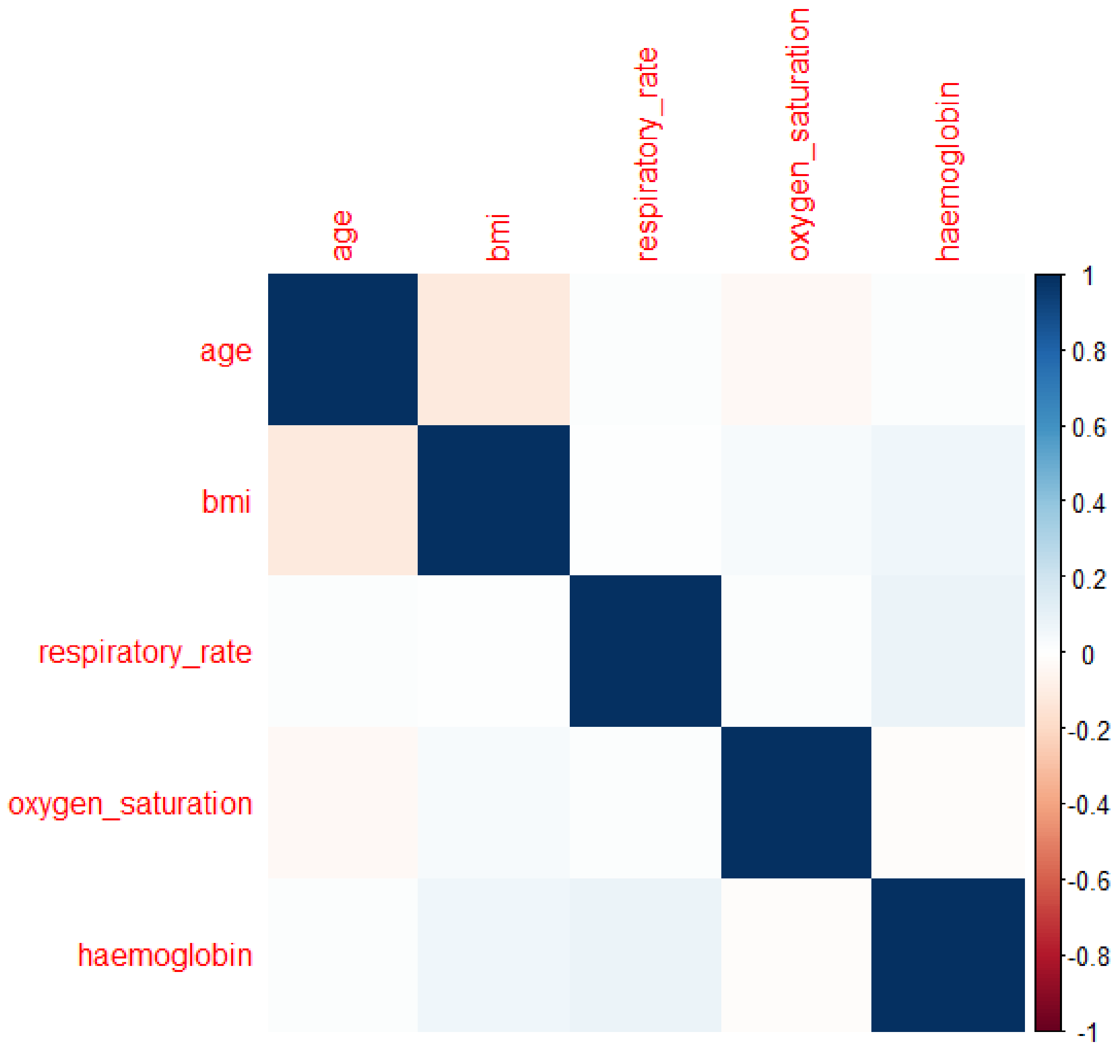



**Interpretation:**

- Moderate Risk patients had the highest median age among all risk categories.
- High Risk and Moderate Risk groups showed greater variability in age compared with the Low Risk group.
- A few older outliers were observed in the Low Risk and NA categories.
- The considerable overlap in age distributions suggests that age alone does not clearly differentiate patients across risk categories.

## Step 5: Correlation Heatmap of Clinical Parameters

```r
vars <- data[, c("age", "bmi", "respiratory_rate", "oxygen_saturation", "haemoglobin")]
corrplot(cor(vars, use = "complete.obs"), method = "color")
```




**Interpretation:**

- Age shows a very weak positive correlation with BMI.
- Age has almost no correlation with respiratory rate, oxygen saturation, or haemoglobin.
- BMI also shows very weak correlations with the remaining variables.
- Respiratory rate, oxygen saturation, and haemoglobin do not show strong relationships with one another.
- Overall, none of the variables show a strong correlation (no dark blue or dark red cells outside the diagonal).

## Summary and Conclusion

Overall, this analysis suggests that age skews right rather than following a normal distribution, BMI is the most commonly abnormal clinical parameter in this population, and — most importantly for the study's core question — the cross-tabulation in Question 8 highlights specific instances where medical officers' subjective risk classification diverged from the objective, parameter-count-based risk category. This supports further exploration of an algorithmic approach as a potential safety net for identifying seriously ill TB patients who might otherwise be under-triaged.